

Algebra 1

Unit 7

Lesson 8 Homework

Name Answer key

Date _____

Period _____

For questions 1-2, consider the expression $16m^2 - 100p^2$.

1. Rewrite the expression as a difference of two squares.

$$(4m)^2 - (10p)^2$$

2. Factor the expression.

$$(4m + 10p)(4m - 10p)$$

For questions 3-4, consider the expression $144a^2 - 49b^2$.

3. Rewrite the expression as a difference of two squares.

$$(12a)^2 - (7b)^2$$

4. Factor the expression.

$$(12a + 7b)(12a - 7b)$$

For questions 5-8, determine whether the quadratic expression is a difference of two squares. Then, factor the expression.

5. $64d^2 - 121g^2$

Yes.

$(8d + 11g)(8d - 11g)$

6. $9f^2 + 48f + 64$

Not the difference of two squares.

$(3f + 8)^2$

7. $1 - r^2$

Yes.

$(1 + r)(1 - r)$

8. $-121 + t^2$

Yes.

$(t + 11)(t - 11)$

For questions 9-10, factor the expression.

9. $x^2 - 2x + 1$

$(x - 1)^2$

10. $16y^2 - 25x^2$

$(4y + 5x)(4y - 5x)$